

Delivering Solutions for the Digital Age

Mike Inglis
EVP Marketing

The Digital Age Has Arrived



My Reality



+



Sharing of Data

- Audio at home, on a plane, in the car, in the house
- Pictures on the PC, on the camera, on the laptop, on the phone
- PCs for communication, analysis and pipes
- Digital TVs for HD, flat screens, display and quality
- To allow our networks to flourish and have meaning



But it's Not Just Technology



ABOUT (PRODUCT) RED AND MOTOROLA

(PRODUCT) RED was created by Bono of U2, and Bobby Shriver, Chairman of DATA, to raise awareness and money for The Global Fund by teaming up with the world's most iconic brands to produce (RED) branded products. A percentage of each (RED) product sold is given to The Global Fund to invest in African AIDS programs with an emphasis on women and children.

Motorola is the exclusive wireless partner for (RED), and in tribute to our involvement we've produced a special (RED) version of the SLVR – the new red MOTOSLVR™.

By purchasing the new red MOTOSLVR™, you've not only got the ultimate mobile phone, you've also allowed us to make a direct contribution – £10 from every sale – to The Global Fund to help eliminate AIDS in Africa. Your participation in the program doesn't end at purchase, but continues every time you use this red MOTOSLVR™ phone; 5% of your normal charges from all calls and texts will be contributed to The Global Fund.

joinred.com



14 'year old' decision in June 2006

- Its Cool and Red
- Ethical donation to Africa
- Pay as you go



Image and brand

Demographics

Price point

What 'We' Are Talking About...

Volume process and capacity

- “90nm technology is the world’s most popular process for IC designs” Synopsys User Group
- “IDMs shift to foundries undermines start ups”

Functionality

- “In a time when the single chip cell phone is on everyone's wish list” David Carey, Portelligent
- “Voice remains the Killer App but what other types of functionality are we capable of absorbing in that (mobile) device – Gaming”... Sanjay Jha, CEO, Qualcomm CDMA Technologies

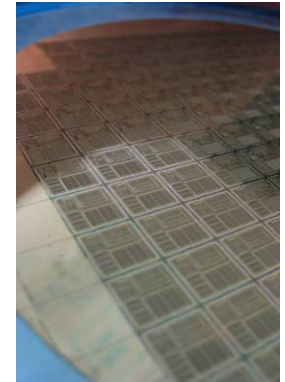
EETIMES



In EETimes...

Advanced technologies

- “TI adds Chartered and may equip new fab for 45nm” Kevin Ritchie, TI
- “TSMC lifts the lid on DFM ecosystem” Ed Wan, TSMC



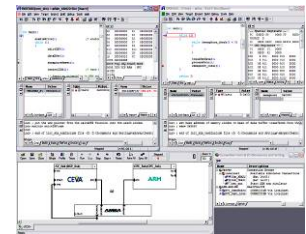
China

- “We’ve had a doubling of the number of consumers in the world” W.Corrigan, LSI



Tools

- “Multicore design strives for balance but debug tools complicates” Richard Goering
- “Eclipse is becoming a widely-used platform for embedded tools” Robert Day

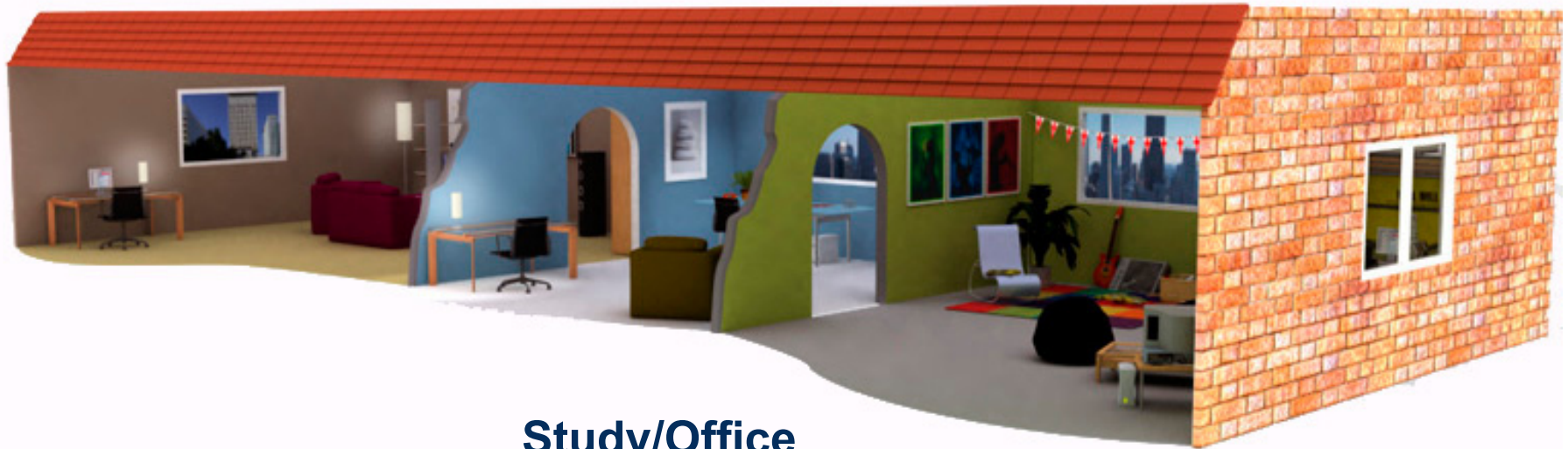


Tools, Technology, Capacity

- Enabling content delivery
- More functionality
 - More integration
 - At the right price
 - In capacity
 - With time-to-market
 - To meet consumer demand
 - Combined with the Killer App - voice !



Greg and Tiff's APM House



Lounge/Sitting Room

- Plasma TV
- Hand controller
- Telephones
- Digital photo frame
- Sound Stereo system/internet radio
- Other domestic appliances

Study/Office

- Office equipment
- Printer/plotter/calculator
- Storage/USB key/router
- Camcorder/photo printer

Bedroom

- Games console
- PC on desk/network camera
- PSP/DS lite/iPod boom box
- Sports watch/robot

Physical IP Die and Processors

ARM Processors

- ARM Powered processors shipped during the APM

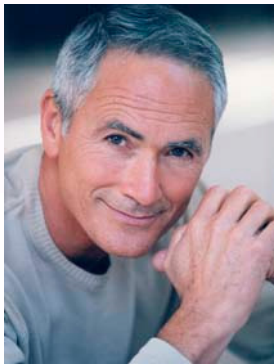
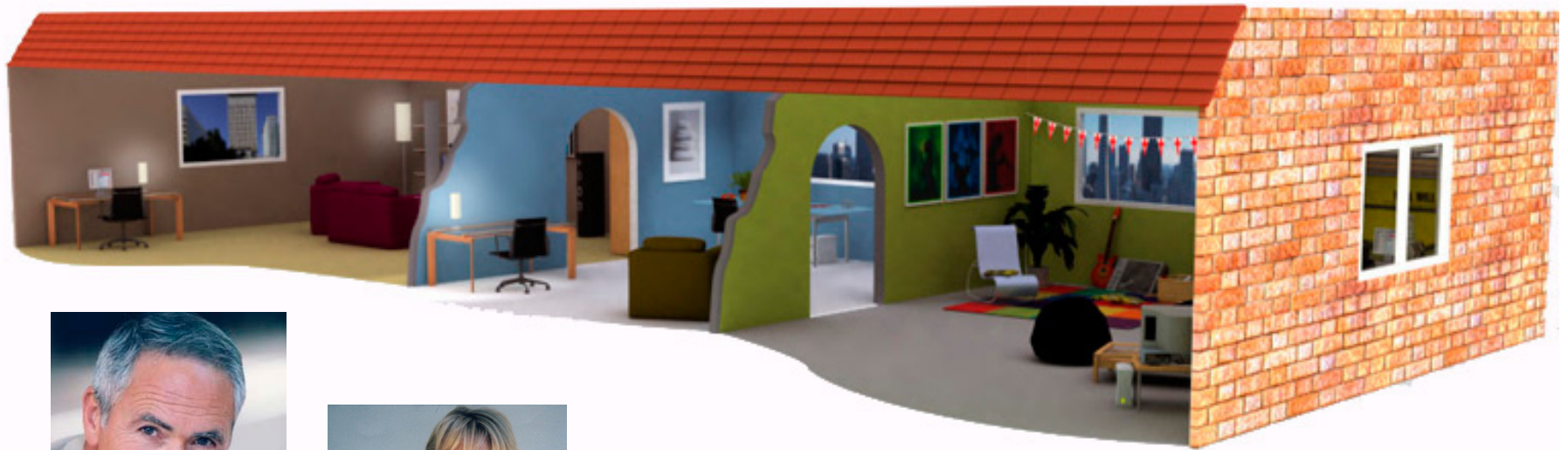


Die with ARM Physical IP

- ARM Powered die shipped during the APM

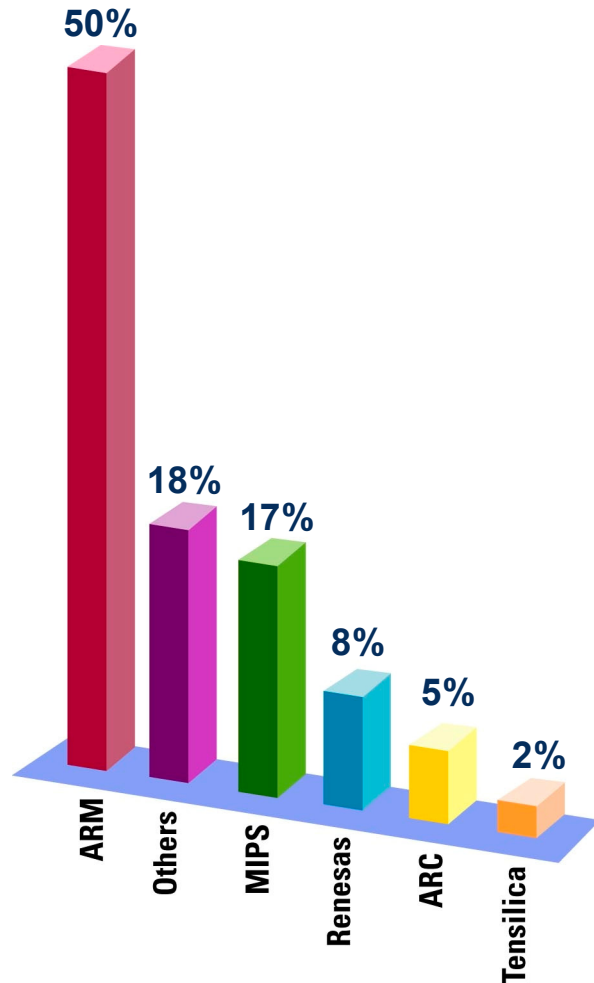


Link to House



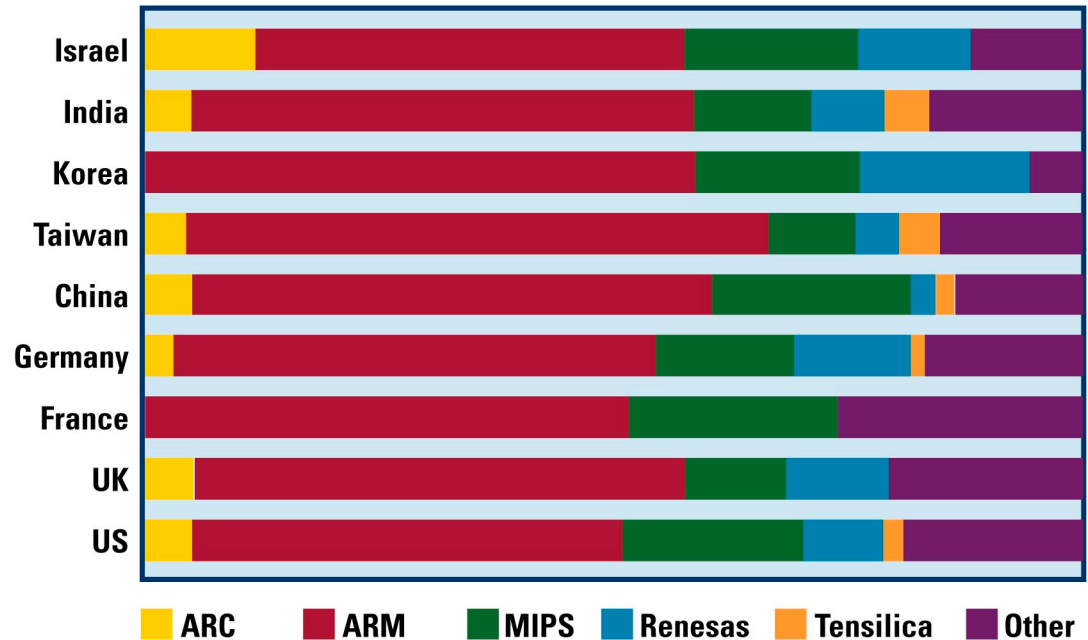
Which Architecture?

Global responses



*Total responses exceed 100% as some respondents are currently using more than one brand of RISC Microprocessor

Response by region



Source: ARM 2006 Global Brand Awareness Study n=978

System and System Cost are Key

2000	2002	2005	2006
Documentation	Component availability	Performance	System cost
Outstanding products	Quality and availability of development tools	32-bit architecture	Development or verification tools
Application support	Quality and availability of technical documentation	Low total system cost	Licensing cost
Competitive price	Component cost	OS support	Processor performance range
After-sales support	Speed (system performance)	Design enabling high reuse	OS support

Sources:

ARM 2006 Global Brand Awareness Study - Townsend

ARM 2005 Global Brand Awareness Study – Context Analytics

ARM 2004 Global Brand Awareness Study – Context Analytics

ARM 2003 Global Brand Awareness Study – CMP/EETimes/Embedded Systems

ARM 2002 Global Brand Awareness Study – Townsend/CMP/EETimes/Embedded Systems

ARM 2000 Global Brand Awareness Study – CMP/EETimes/Embedded Systems

2006 Brand Survey n = 978

What Does ARM Do?

A partnership with our customers and broader community of third parties, to enable the creation of end products more efficiently through ARM than from any other source



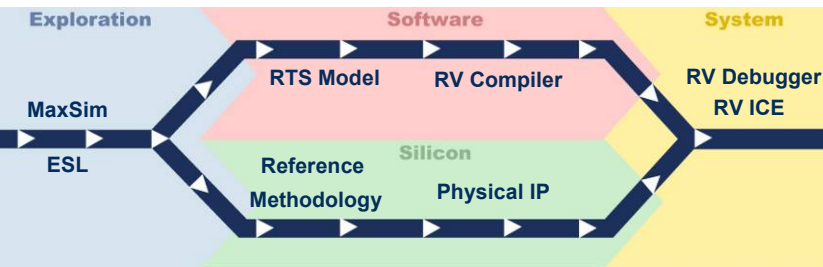
Software Ecosystems and Hardware



TrustZone™ Software IEM Software Execution Environment Swerve 3D graphics

ARM Processor AudioDE™ MBX AMBA Designer Memory Controller

Power Management Kit Libraries Memory PHYs



ARM Announces Availability of Mobile Consumer DRM Software Solutions Based on ARM TrustZone Technology

CAMBRIDGE, UK – Feb. 14, 2006 – ARM (LSE: ARM); (Nasdaq: ARMHY) today announced at the 3GSM World Congress, Barcelona, that ARM, Obigo and Philips Software will demonstrate Open Mobile Alliance (OMA) technology-compliant DRM features based on the [ARM® TrustZone® Software](#) framework in upcoming mobile software offerings.

The TrustZone Software API provides a common framework for secure application and device developers, reducing the time and cost of developing secure applications.

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Press Release

Source: Texas Instruments Incorporated

Texas Instruments and ARM Accelerate Deployment of Secure Applications on Mobile Devices

Tuesday April 4, 5:00 am ET

Market [Semiconductor > News > SystemLSI](#)

Phone (Mar 21, 2006 / SEC)

Samsung Develops Enhanced Multimedia Processor for Mobile Applications

DALLAS (NYSE: [TXI](#)) announced today enabling next-generation application technology.



Taipei, Taiwan - March 21, 2006 : Samsung Electronics Co., Ltd., the leader in advanced semiconductor technology, announced a new mobile AP for enhanced multimedia performance at the third annual Samsung Mobile Solution Forum. The new processor is the ultimate solution for multimedia intensive mobile handsets.

Samsung's new multimedia mobile AP covers a broad scale of powerful multimedia features. It incorporates multi-format CODEC as dedicated

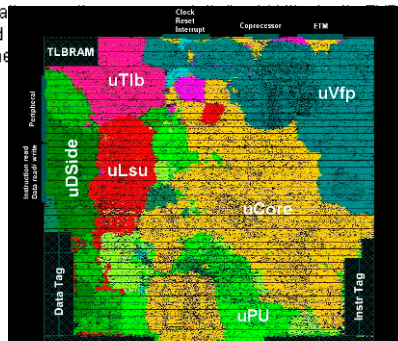


Processors, Tools and Physical IP

ARM Processor and Physical IP deliver groundbreaking 750 MHz ARM11 Implementation

CAMBRIDGE, UK - Jan. 25 2006 – ARM [(LSE: ARM); (Nasdaq: ARMHY)] today unveiled a ground high performance implementation of the ARM1176JZF-S™ processor, enhanced with the ARM Advantage™ cell library and memories, which achieves a frequency of more than 750 MHz in a 1 performance 90nm foundry process while occupying only 2.4 mm² of silicon area. This represents significant performance increase through a combination of collaborative design, advanced process technology. The optimized implementation delivers an industry-leading performance of more than 920 Dhrystone MIPS for existing applications and operating systems without the need for software re-design or re-compilation.

"The industry leading combination of the ARM processor business and the Physical IP business from Artisan a year ago is delivering technical and Business Development." Our combined class 90nm design for semiconductor Partners



ARM1176JZF-S	90nm TSMC90GT
Die Area including 16K I & D caches	2.4 mm ²
Performance NVT	+750MHz (ss, 1.08V, +125C)
Benchmarking	922 DMIPS
Total Power including leakage	< 500mW (tt, 1.2V, +25C)



Device Database
List of Supported Chips
Parametric Chip Search
List Chips by Architecture
List Chips by Vendor
ID Codes
What is the Device Data?
Report Problems
Tell Us What You Think

RealView Microcontroller Development Kit
Supports ARM7, ARM9, Cortex-M3 based devices

The RealView Microcontroller Development Kit (MDK) shortens development cycles by reducing time spent configuring testing and debugging embedded applications. The RealView MDK combines the ARM RealView compilation tools with the Keil uVision Integrated Development Environment (IDE), providing developers with a feature-rich environment optimized for ARM Powered microcontrollers.

The Keil uVision IDE includes:

- Project Management and Device & Tool Configuration.
- Source Code Editor Optimized for Embedded Systems.
- Target Debugging and Flash Programming.
- Accurate Device Simulation (CPU and Peripheral).

ARM technology-based projects created under uVision are automatically compiled and linked using the RealView compilation tools.

The bulk microcontroller simulator models more than 50 ARM Powered devices including the ARM instruction set, the on-chip peripherals, and the external signals used to manipulate them.

Logic Analyzer
Waveform showing digital signals over time.



Solutions and Standards

Cortex

AudioDE™

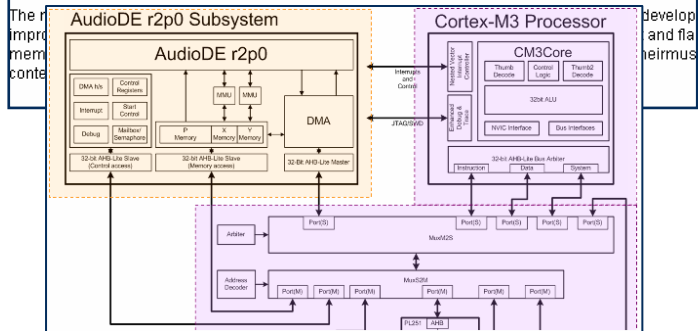
AMBA³
System Architecture

RealView®
Tools by ARM®

ARM Brings Best-in-Class Audio Solutions to Portable Consumer Devices

OptimoDE data processing technology enables ultra power efficient, multi-standard digital audio codec support for devices such as MP3 players

CAMBRIDGE, UK – May 16 2005—ARM (LSE:ARM); (Nasdaq:ARMHY) announced today at the Spring Processor Forum in San Jose, Calif., the ARM® OptimoDE™ Audio Edition embedded data processing technology optimized for the efficient implementation of digital audio in embedded devices.



Item	Area (mm ²)	Power (mW)
AudioDE and subsystem	0.62	1.12
Memory required to run mp3		
Program memory 20KB (64-bit wide)	0.52	0.34
X-Data memory 12KB (32-bit wide)	0.34	0.11
Y-Data memory 6KB (32-bit wide)	0.19	0.01
AudioDE, subsystem and memory	1.67	1.58
Cortex-M3, bus and peripherals	0.55	0.62
Totals	2.22	2.20



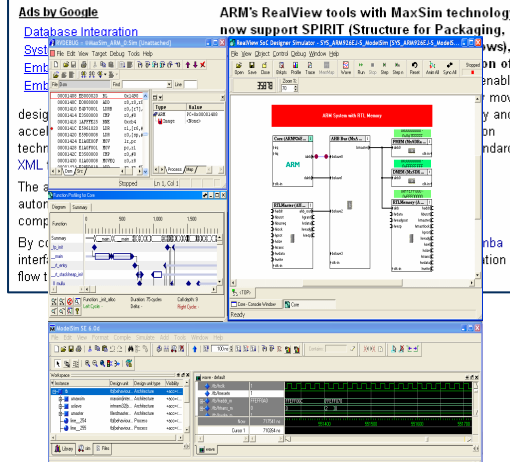
EDA Tech Forum® >> REGISTER
Technical Resource for the EE Community
FREE, one day technical seminar
Wantage, UK - June 16
RBS Williams F1 Circuit

News Release from: ARM
Edited by the Electronicstalk Editorial Team on 16 June 2005

SoC design tools support SPIRIT standard

ARM's RealView tools with MaxSim technology now support SPIRIT, the industry standard for efficient integration of IP in SoC platform design.

Note: Readers of the Editor's free email newsletter will have read this news when it was announced. [Send us a blank email now to join the circulation.](#) It's free!



cadence

carbon
design systems

ESTEREL
Technologies

The MathWorks

Mentor
Graphics

SpiralTech

SYNOPSYS®

Tenison
Design Automation

The SPIRIT
Consortium

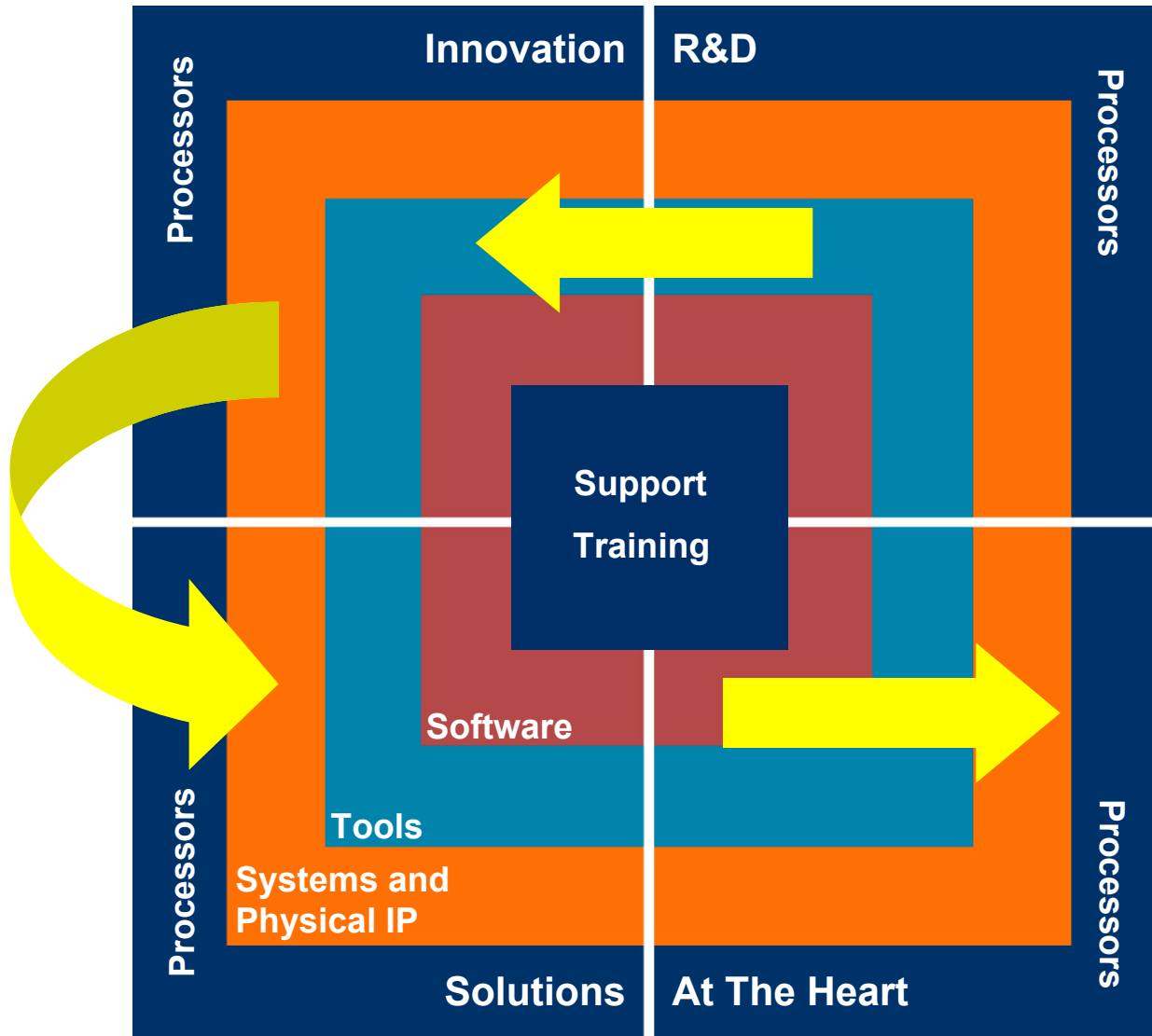
ARM
partner meeting
2006

THE ARCHITECTURE FOR THE DIGITAL WORLD®

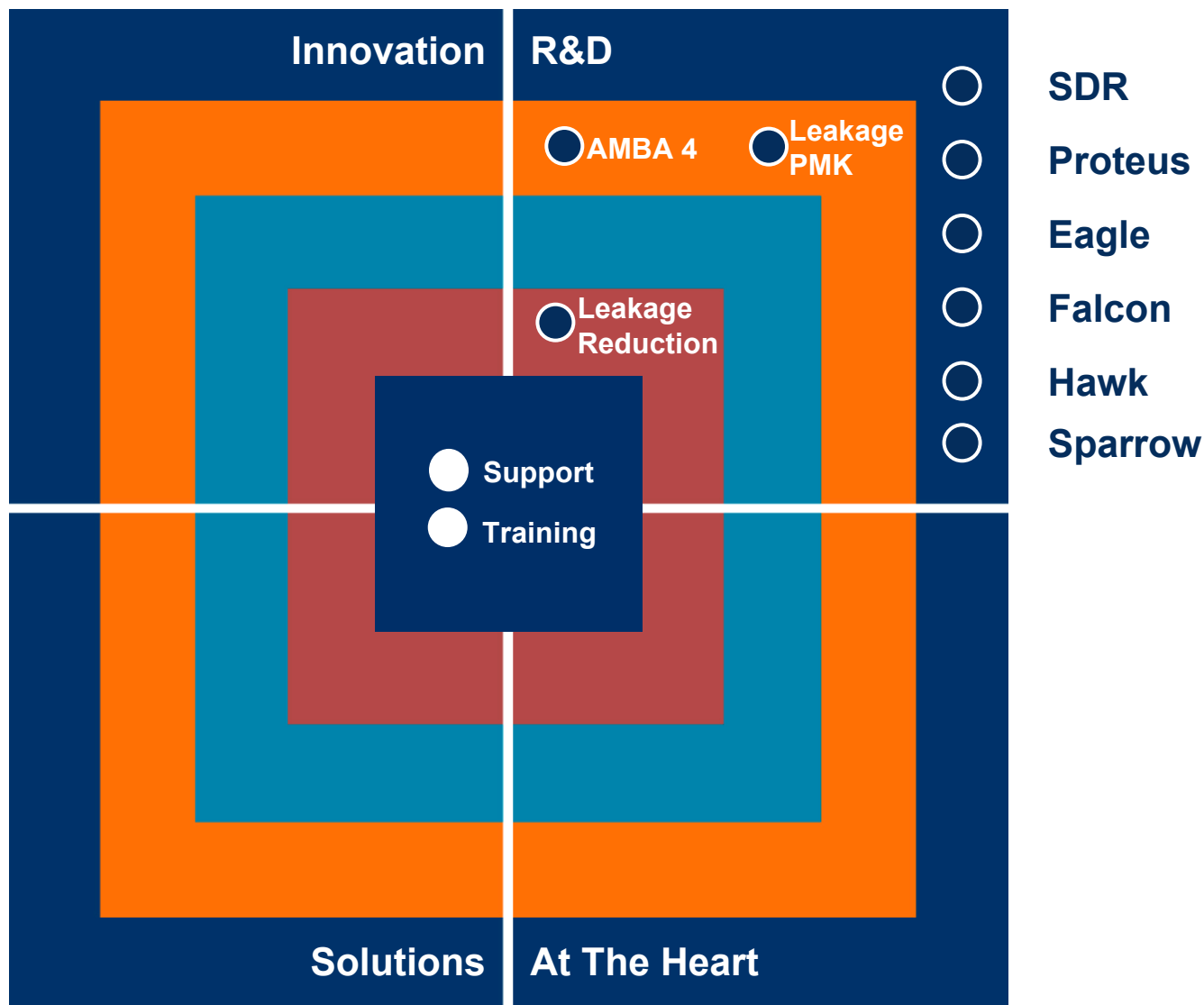
Confidential 17

ARM®

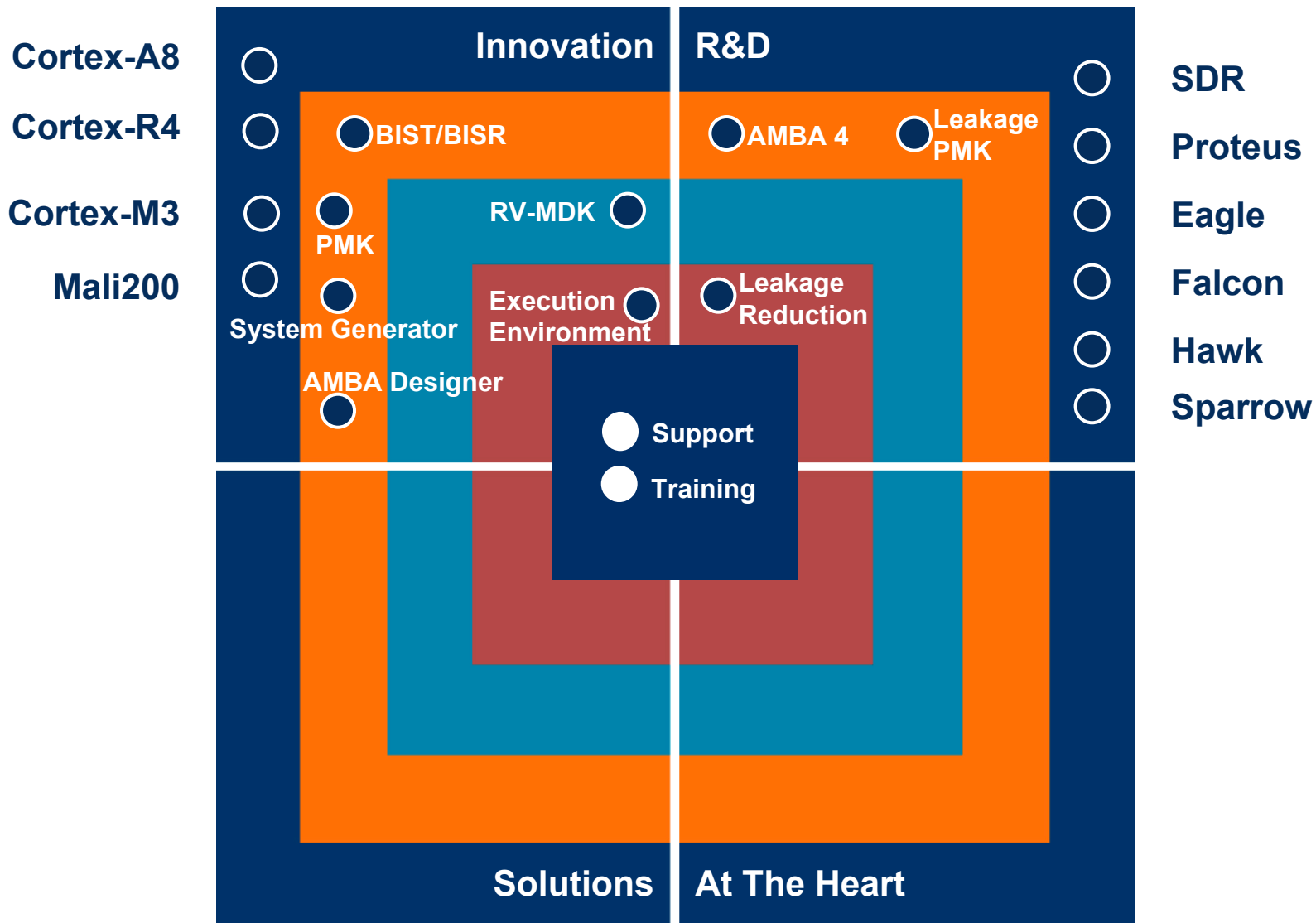
Innovation in Time



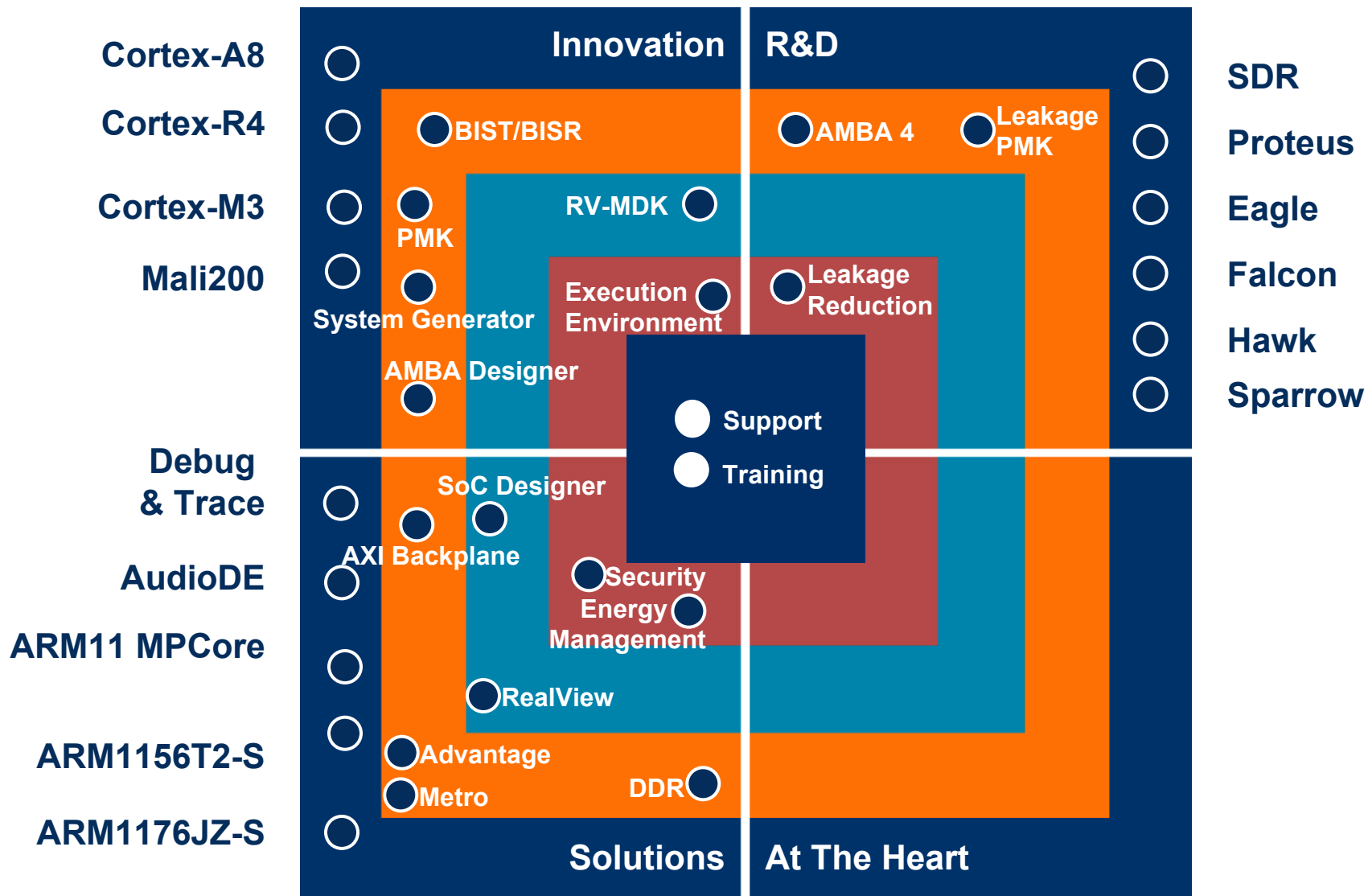
R&D



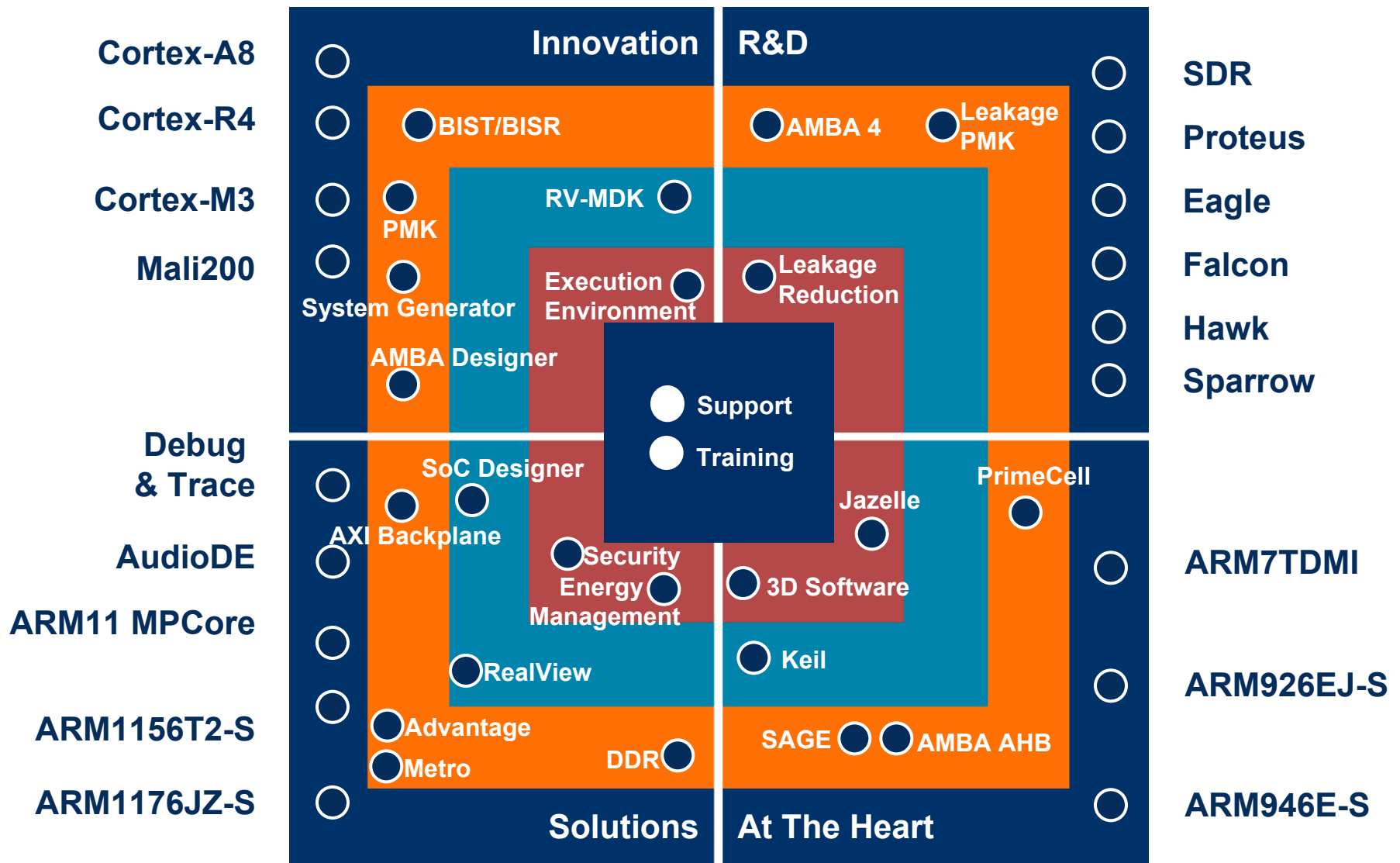
Innovation



Solutions

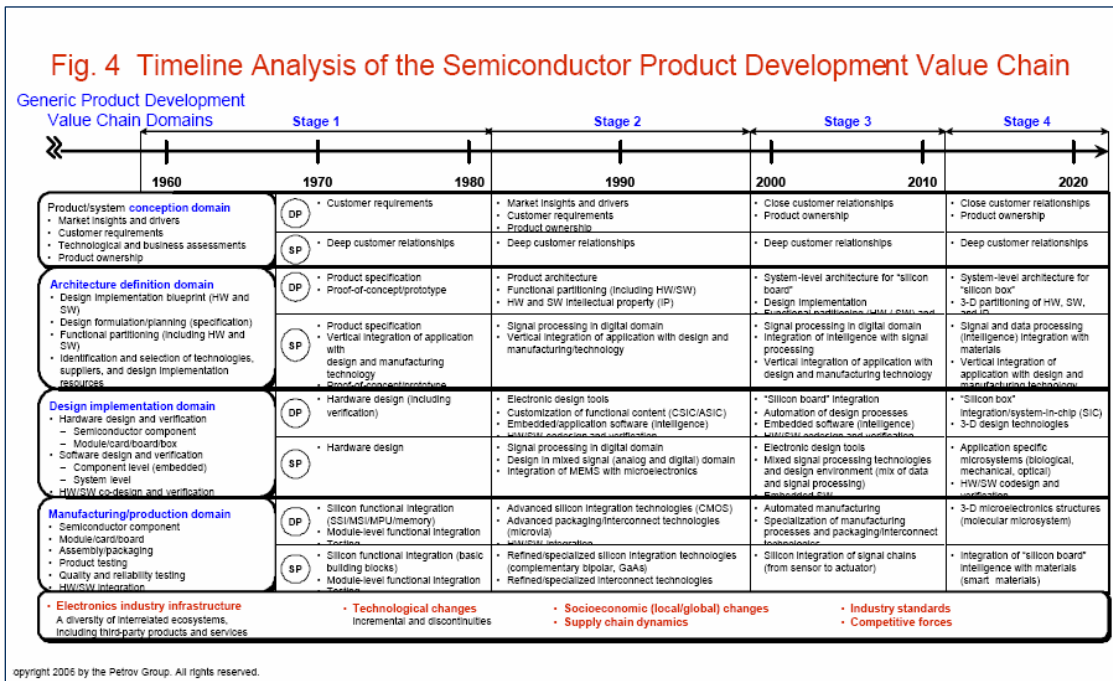


At The Heart



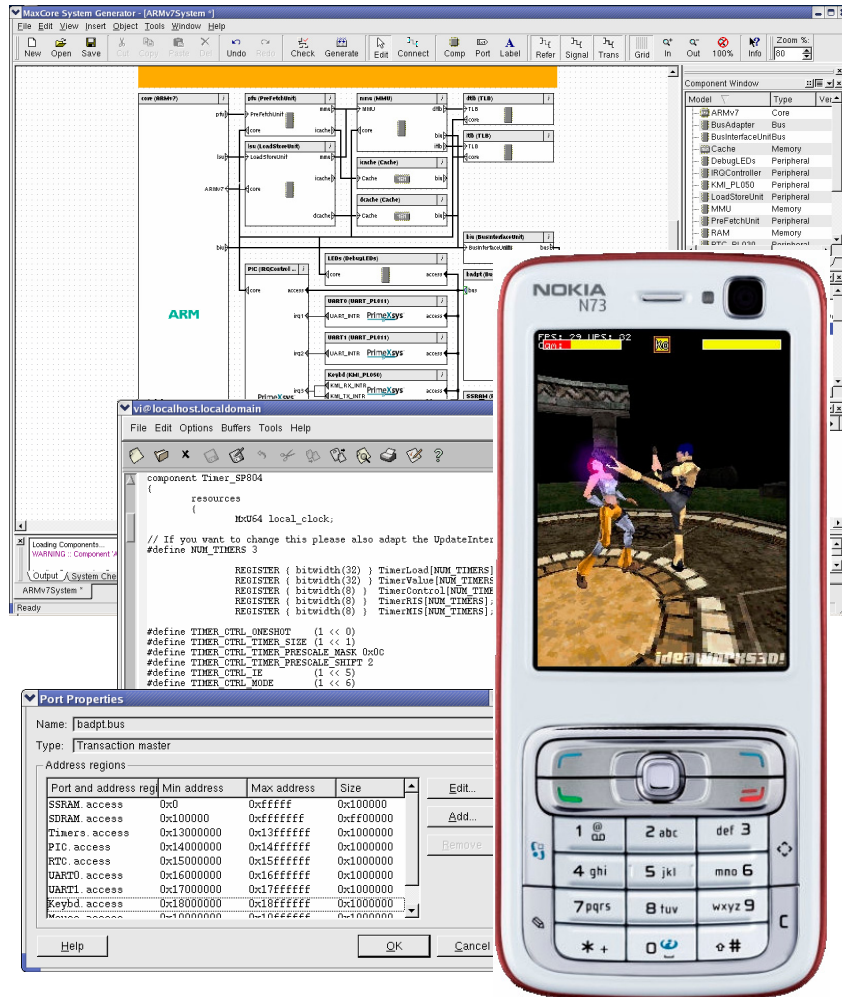
Software: More Mission Critical in 2015+

- “Software: A gradual and predictable increase in levels of software is driven by Moore’s Law”
- “Viewed from a higher plane, software in data processing embodies the user’s application”



Source: Petrov Group
May 2006

Changing How We Design

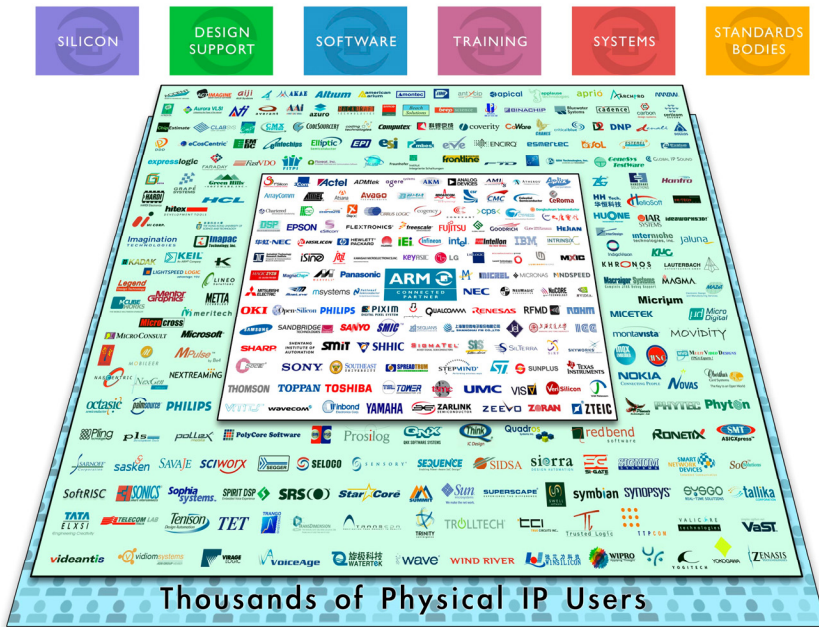


- Hardware platforms not available early enough for content development
- System Generator produces very fast models of ARM processor-based systems
 - 100-200MHz speed on PC
 - ARM-certified processor models
- Develop model of your future chip or end-user device for multiple uses

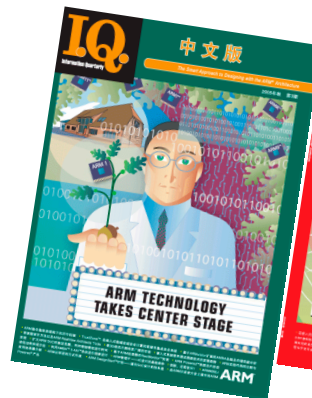
ideaworks3D!TM

RealView[®]
Tools by ARM[®]

The Value is in Your Network



- Ability to pitch articles in ***IQ*** (65K global circulation) or advertise
- Chance to exhibit or sponsor at the ***ARM Developers' Conference***
- Quarterly ***e-newsletter*** to members about new opportunities for members



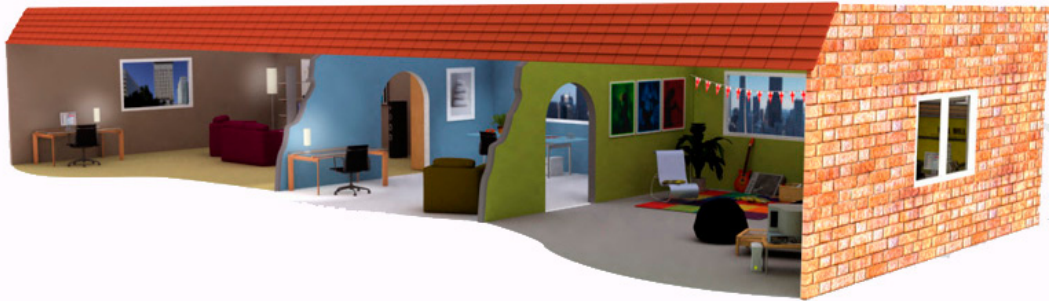
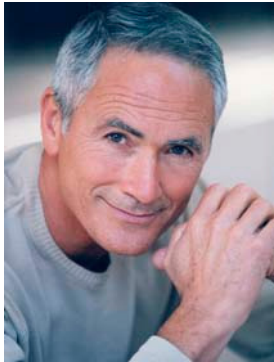
Connected Community Q4 Events

- ARM Developers' Conference
 - 03 Oct-05 Oct *Santa Clara, USA*
- ARM Connected Community Technical Symposiums
 - 17 Oct *Tokyo, Japan*
 - 06 Nov *New Delhi*
 - 08 Nov *Hyderabad, India*
 - 10 Nov *Bangalore, India*
 - 14 Nov *Herzlyia, Israel*
 - 16 Nov *Paris, France*
 - 28 Nov *Seoul, Korea*
 - 30 Nov *HsinChu, Taiwan*
 - 04 Dec *Shanghai, China*
 - 06 Dec *Beijing, China*
 - 08 Dec *Shenzhen, China*



Summary

- The Digital Age has arrived
- We are focused on solving the whole SoC design problem
- We are innovating in all areas of our product range
- Software is going to get even more critical in 2006, 2010 and 2015+
- Greg and Tiff are in the ARM Powered house



Segment Tracks



HOME
SOLUTIONS



ENTERPRISE
SOLUTIONS



MOBILE
SOLUTIONS



EMBEDDED
SOLUTIONS

EMERGING APPLICATIONS